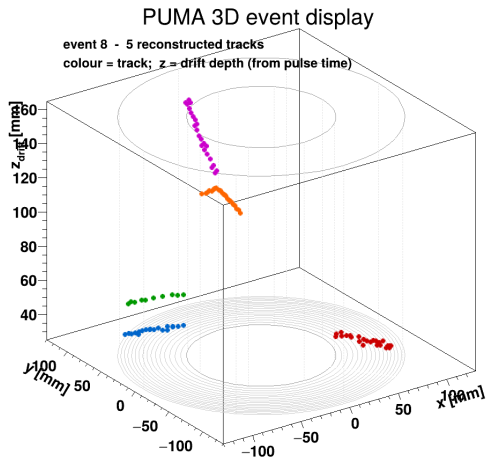


PUMA TPC — track reconstruction

Performance on the $\pi^+\pi^-$ channel

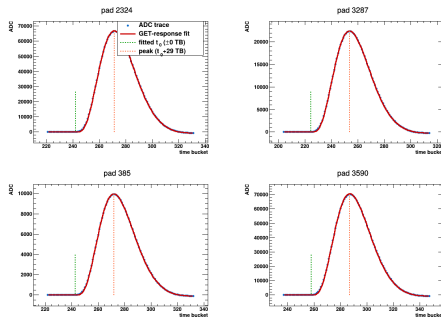
The detector



Annular P10 TPC, 4 T. Pads $\rightarrow (x, y)$; drift time $\rightarrow z$.

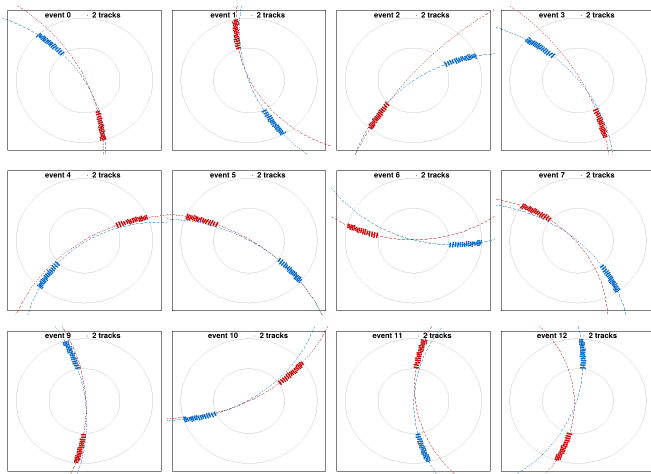
Reconstruction chain

pulse fit $\rightarrow$ pattern recognition $\rightarrow$ Kalman fit



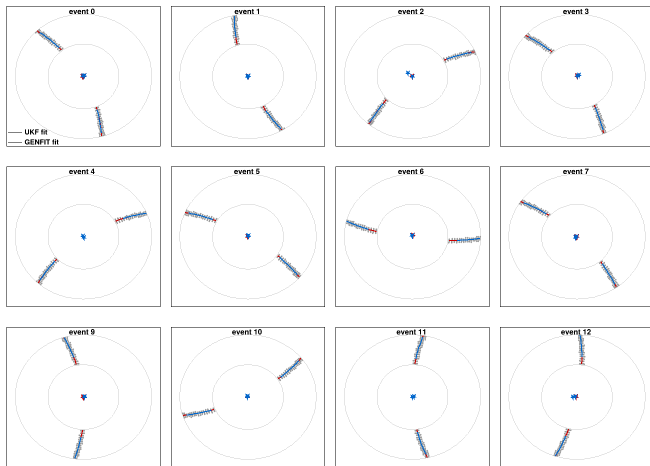
AtPSAMultiFit $\rightarrow$ HDBSCAN $\rightarrow$ UKF & GENFIT.

Pattern recognition



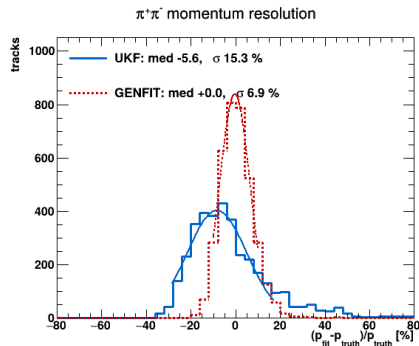
HDBSCAN; tracking efficiency **99.7%**. Hits + circle fit.

Track fitting



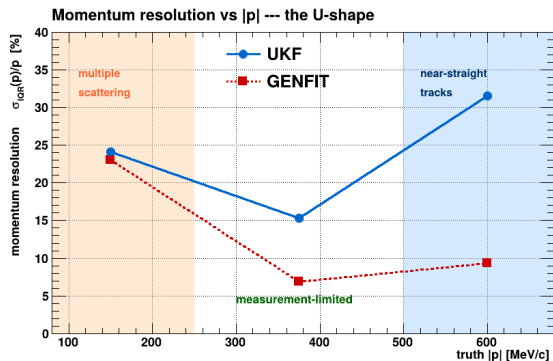
UKF & GENFIT; fitting efficiency $\sim 100\%$.

Momentum resolution

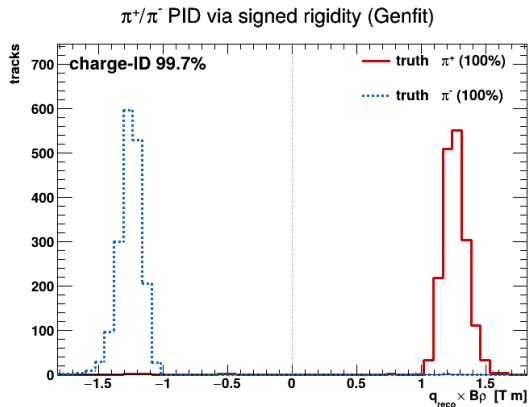


	momentum σ	θ σ	vertex σ_z	charge
UKF	15%	2.0°	0.50 mm	99.5%
GENFIT	7%	1.8°	0.47 mm	99.6%

Resolution vs momentum



U-shape, best at 375–500 MeV/c. Low p : **multiple scattering** ($\propto 1/p\beta$) — an irreducible floor (both fitters converge). High p : **near-straight** tracks, little curvature to measure.



Charge sign from rigidity: **99.7%**, 14.5σ separation.

Summary

tracking efficiency	99.7%
fitting efficiency	$\sim 100\%$
momentum resolution (GENFIT)	7%
polar-angle resolution	1.8°
vertex resolution σ_z	0.5 mm
charge / $\pi^+\pi^-$ ID	99.7%

Full chain runs natively; GENFIT is the stronger fitter.